databricks trans / 1.race_results

(https://databricks.com)

Read all data from all data sources as required

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
# drivers_df = spark.read.parquet(f"{processed_folder_path}/drivers") \
#     .withColumnRenamed("name", "driver_name") \
#     .withColumnRenamed("number", "driver_number") \
#     .withColumnRenamed("nationality", "driver_nationality")
```

```
drivers_df = spark.read.format("delta").load(f"{processed_folder_path}/drivers") \
    .withColumnRenamed("number", "driver_number") \
    .withColumnRenamed("name", "driver_name") \
    .withColumnRenamed("nationality", "driver_nationality")
```

```
# constructors_df = spark.read.parquet(f"{processed_folder_path}/constructors") \
#     .withColumnRenamed("name", "team")
```

```
constructors_df = spark.read.format("delta").load(f"{processed_folder_path}/constructors") \
.withColumnRenamed("name", "team")
```

```
circuits_df = spark.read.format("delta").load(f"{processed_folder_path}/circuits") \
.withColumnRenamed("location", "circuit_location")
```

```
# circuits_df = spark.read.parquet(f"{processed_folder_path}/circuits") \
#     .withColumnRenamed("location", "circuit_location") \
#     .withColumnRenamed("circuit_Id", "circuit_id")
```

```
# races_df = spark.read.parquet(f"{processed_folder_path}/races") \
#     .withColumnRenamed("name", "race_name") \
#     .withColumnRenamed("race_timestamp", "race_date")
```

```
races_df = spark.read.format("delta").load(f"{processed_folder_path}/races") \
.withColumnRenamed("name", "race_name") \
.withColumnRenamed("race_timestamp", "race_date")
```

```
# results_df = spark.read.parquet(f"{processed_folder_path}/results") \
#     .withColumnRenamed("time", "race_time")
```

```
results_df = spark.read.format("delta").load(f"{processed_folder_path}/results") \
.filter(f"file_date = '{v_file_date}'") \
.withColumnRenamed("time", "race_time") \
.withColumnRenamed("race_id", "result_race_id") \
.withColumnRenamed("file_date", "result_file_date")
```

Join circuits to races

```
race_circuits_df = races_df.join(circuits_df, races_df.circuit_id == circuits_df.circuit_Id, "inner") \
    .select(races_df.race_id, races_df.race_year, races_df.race_name, races_df.race_date, circuits_df.circuit_location)
```

```
# circuits_df.show()
```

```
------------------+
|         1|   albert_park|Albert Park Grand...|       Melbourne|Australia|-37.8497|  144.968|      10|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         2|        sepang|Sepang Internatio...|    Kuala Lumpur| Malaysia| 2.76083|  101.738|      18|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         3|       bahrain|Bahrain Internati...|          Sakhir|  Bahrain| 26.0325|  50.5106|       7|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         4|     catalunya|Circuit de Barcel...|        Montmeló|    Spain|   41.57|  2.26111|     109|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         5|      istanbul|       Istanbul Park|        Istanbul|   Turkey| 40.9517|   29.405|     130|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         6|        monaco|   Circuit de Monaco|     Monte-Carlo|   Monaco| 43.7347|  7.42056|       7|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         7|    villeneuve|Circuit Gilles Vi...|        Montreal|   Canada|    45.5| -73.5228|      13|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         8|   magny_cours|Circuit de Nevers...|     Magny Cours|   France| 46.8642|  3.16361|     228|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|         9|   silverstone| Silverstone Circuit|     Silverstone|       UK| 52.0786| -1.01694|     153|     Ergast|2021-03-21|20
24-07-25 06:54:...|
|        10|hockenheimring|      Hockenheimring|      Hockenheim|  Germany| 49.3278|  8.56583|     103|     Ergast|2021-03-21|20
24-07-25 06:54:...|
```

```
# display(race_circuits_df)
```

Table

	race_id	race_year	race_name	race_date	circuit_location
21	1071	2021	Brazilian Grand Prix	2021-11-07T17:00:00.000+00:...	São Paulo
22	1072	2021	Saudi Arabian Grand Prix	2021-12-05T16:00:00.000+00:...	Jeddah

22	1072	2021	Saudi Arabian Grand Prix	2021-12-05T16:00:00.000+00:...	Jeddah
23	1073	2021	Abu Dhabi Grand Prix	2021-12-12T13:00:00.000+00:...	Abu Dhabi
24	989	2018	Australian Grand Prix	2018-03-25T05:10:00.000+00:...	Melbourne
25	990	2018	Bahrain Grand Prix	2018-04-08T15:10:00.000+00:...	Sakhir
26	991	2018	Chinese Grand Prix	2018-04-15T06:10:00.000+00:...	Shanghai
27	992	2018	Azerbaijan Grand Prix	2018-04-29T12:10:00.000+00:...	Baku
28	993	2018	Spanish Grand Prix	2018-05-13T13:10:00.000+00:...	Montmeló
29	994	2018	Monaco Grand Prix	2018-05-27T13:10:00.000+00:...	Monte-Carlo
30	995	2018	Canadian Grand Prix	2018-06-10T18:10:00.000+00:...	Montreal
31	996	2018	French Grand Prix	2018-06-24T14:10:00.000+00:...	Le Castellet
32	997	2018	Austrian Grand Prix	2018-07-01T13:10:00.000+00:...	Spielburg
33	998	2018	British Grand Prix	2018-07-08T13:10:00.000+00:...	Silverstone
34	999	2018	German Grand Prix	2018-07-22T13:10:00.000+00:...	Hockenheim
35	1000	2018	Hungarian Grand Prix	2018-07-29T13:10:00.000+00:...	Budapest
36	1001	2018	Belgian Grand Prix	2018-08-26T13:10:00.000+00:...	Spa

1,058 rows

Now join the above dataframe with other dataframes

```
race_results_df = results_df.join(race_circuits_df, results_df.result_race_id == race_circuits_df.race_id) \
                    .join(drivers_df, drivers_df.driver_id == results_df.driver_id) \
                    .join(constructors_df, constructors_df.constructor_id == results_df.constructor_id)
```

```
from pyspark.sql.functions import current_timestamp
```

```
final_df = race_results_df.select("race_id", "race_year", "race_name", "race_date", "circuit_location", "driver_name", "driver_numb
                                  "team", "grid", "fastest_lap", "race_time", "points", "position", "result_file_date") \
                          .withColumn("created_date", current_timestamp()) \
                          .withColumnRenamed("result_file_date", "file_date")
```

```
# display(final_df.filter("race_year == 2020 and race_name == 'Abu Dhabi Grand Prix'").orderBy(final_df.points.desc()))
```

```
# final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_presentation.race_results")
```

```
merge_condition = "tgt.driver_name = src.driver_name AND tgt.race_id = src.race_id"
merge_delta_data(final_df, 'f1_presentation', 'race_results', presentation_folder_path, merge_condition, 'race_id')
```

```
# %sql
# select * from f1_presentation.race_results;
```

Table

	race_id	race_year	race_name	race_date	circuit_location	driver_name	dr
1	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Charles Pic	

2	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Jean-Eric Vergne	
3	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Daniel Ricciardo	
4	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Pastor Maldonado	
5	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Nico Hülkenberg	
6	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Romain Grosjean	
7	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Pedro de la Rosa	
8	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Sebastian Vettel	
9	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Jenson Button	
10	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Mark Webber	
11	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Felipe Massa	
12	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Kimi Räikkönen	
13	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Nico Rosberg	
14	873	2012	Singapore Grand P...	2012-09-23T12:00:00.000+00:...	Marina Bay	Lewis Hamilton	

10,000+ rows | Truncated data due to row limit

Table

	race_id	count(1)
21	1029	20
22	1028	20
23	1027	20
24	1026	20
25	1025	20
26	1024	20
27	1023	20
28	1022	20
29	1021	20
30	1020	20
31	1019	20
32	1018	20
33	1017	20
34	1016	20
35	1015	20
36	1014	20

1,037 rows